Analysis of technology trends and Developer demographics

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EXECUTIVE SUMMARY

- This analysis was conducted using survey_data_updated.csv, and was visualized using IBM Cognos analytics.
- A few widely used programming languages, databases, platforms, and web frameworks dominate the use of technology today.
- Future technological trends point to a shift toward in demand and cloud-oriented solutions as interest in modern and future technologies increases.
- The majority of responders are in the early to mid-career stages, and the response population is diverse in terms of age groups, regions, and levels of education.

INTRODUCTION

- The purpose of this analysis is to use IBM Cognos Analytics to evaluate survey data in order to determine significant demographic trends among respondents, future technological trends, and current technology usage.
- Students, teachers, and entry-level data and technology professionals looking for information on new trends and technology adoption are the target audience for this research.
- The analysis offers data-driven insights that assist with informed learning and career planning decisions, help stakeholders understand current industry practices, and predict future skill demands.

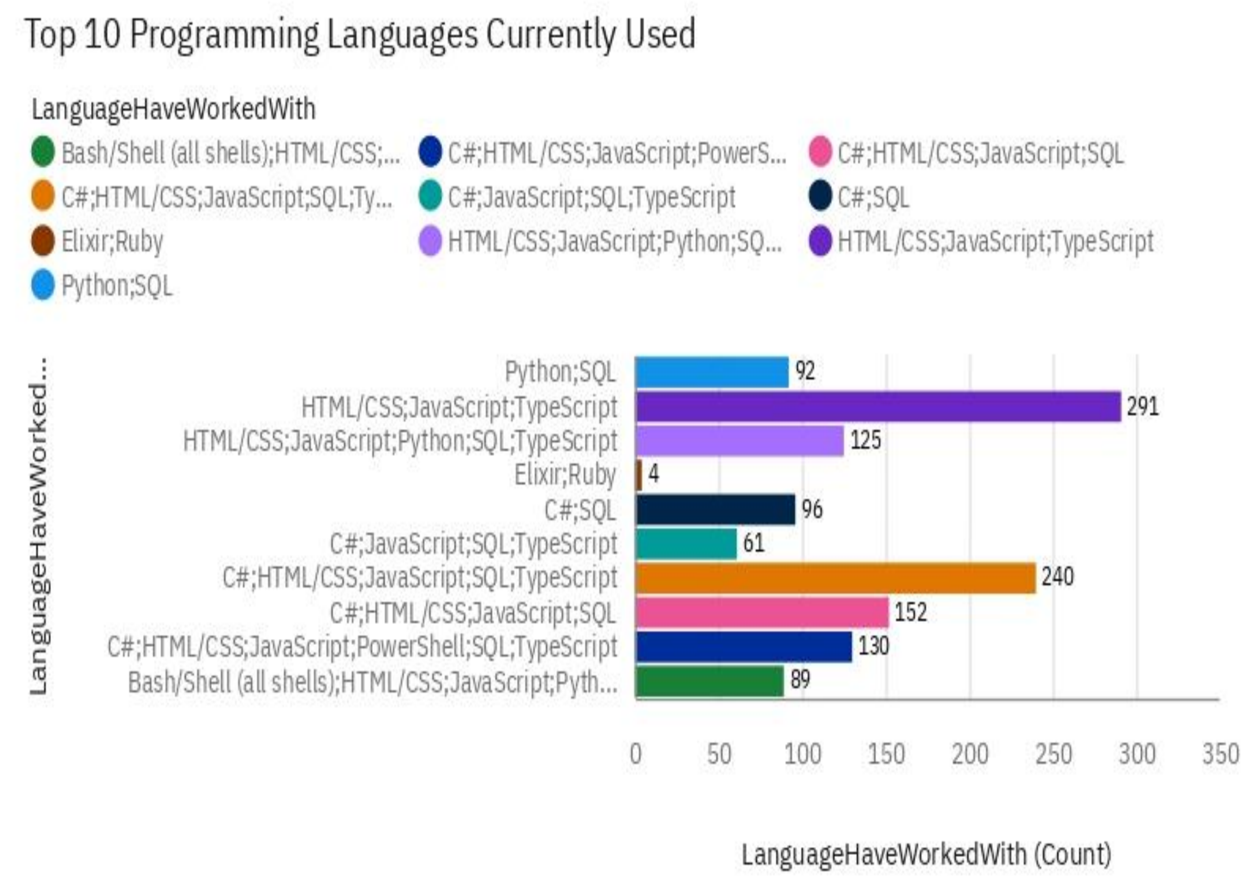
METHODOLOGY

- The survey_data_updated.csv dataset, which includes data gathered from a structured technology survey, is used in the analysis.
- Web scraping and API access were used to gather data, which allowed for the extraction of extensive, structured survey data from online platforms.
- Data preparation included identifying and removing duplicate records, detecting and imputing missing values, and normalizing data to ensure consistency and accuracy.

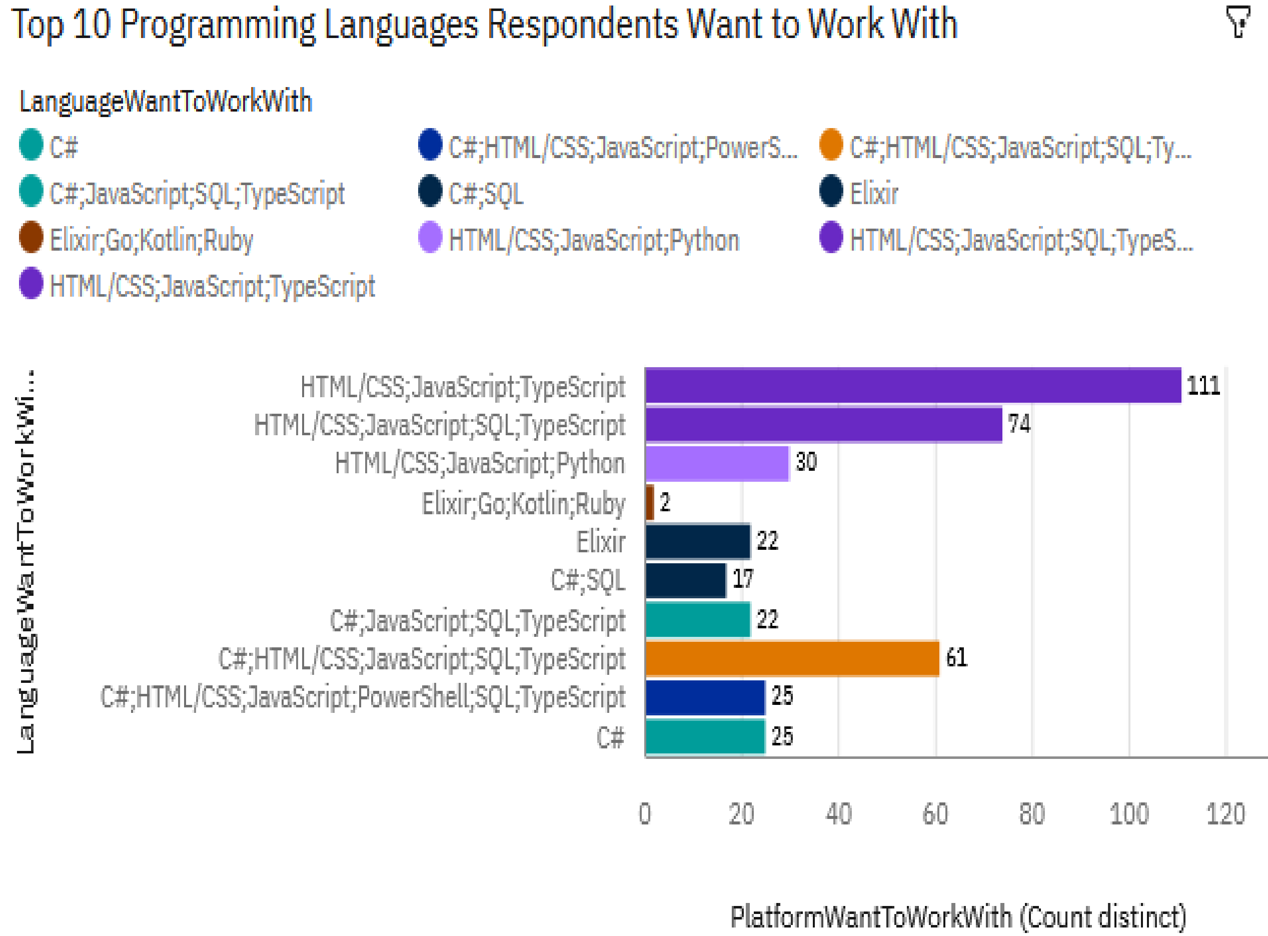
PROGRAMMING LANGUAGE TRENDS

Current Year

Current Technology Usage



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Current usage is dominated by a small handful of well-known programming languages, which reflects industry demand for dependable and extensively supported technology.
- The adoption of programming languages nowadays exhibits consistent usage patterns among responders, suggesting standardization in development procedures.
- Future use of new programming languages is more popular than present use, indicating a predicted change in technological preferences.

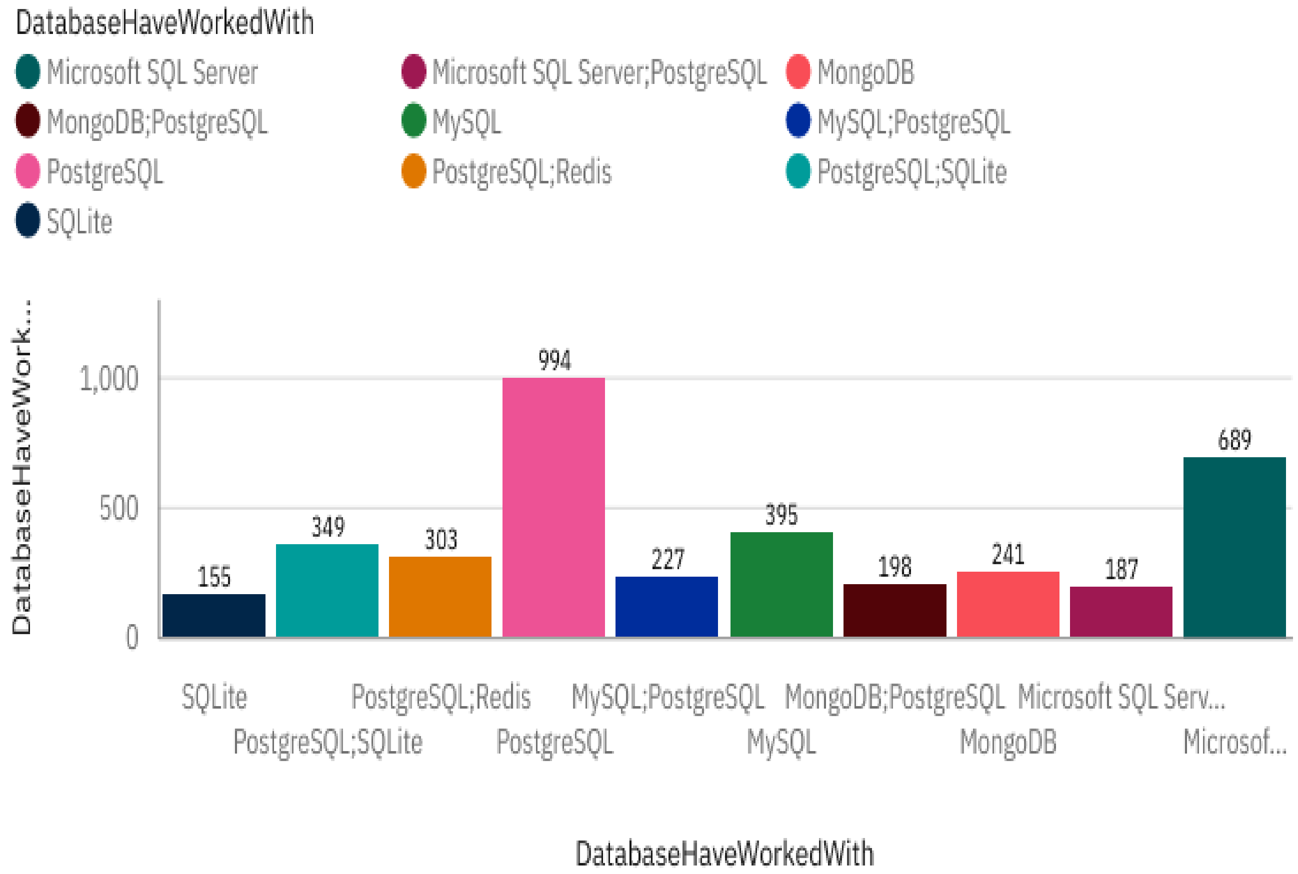
Implications

- Professionals and learners should gradually develop skills in emerging languages while focusing on core, commonly used languages.
- As language preferences change, organizations may need to plan for training and retraining initiatives to stay competitive.
- The materials used in educational programs can be modified to ensure a balance between future developments in technology and the demands of industry.

DATABASE TRENDS

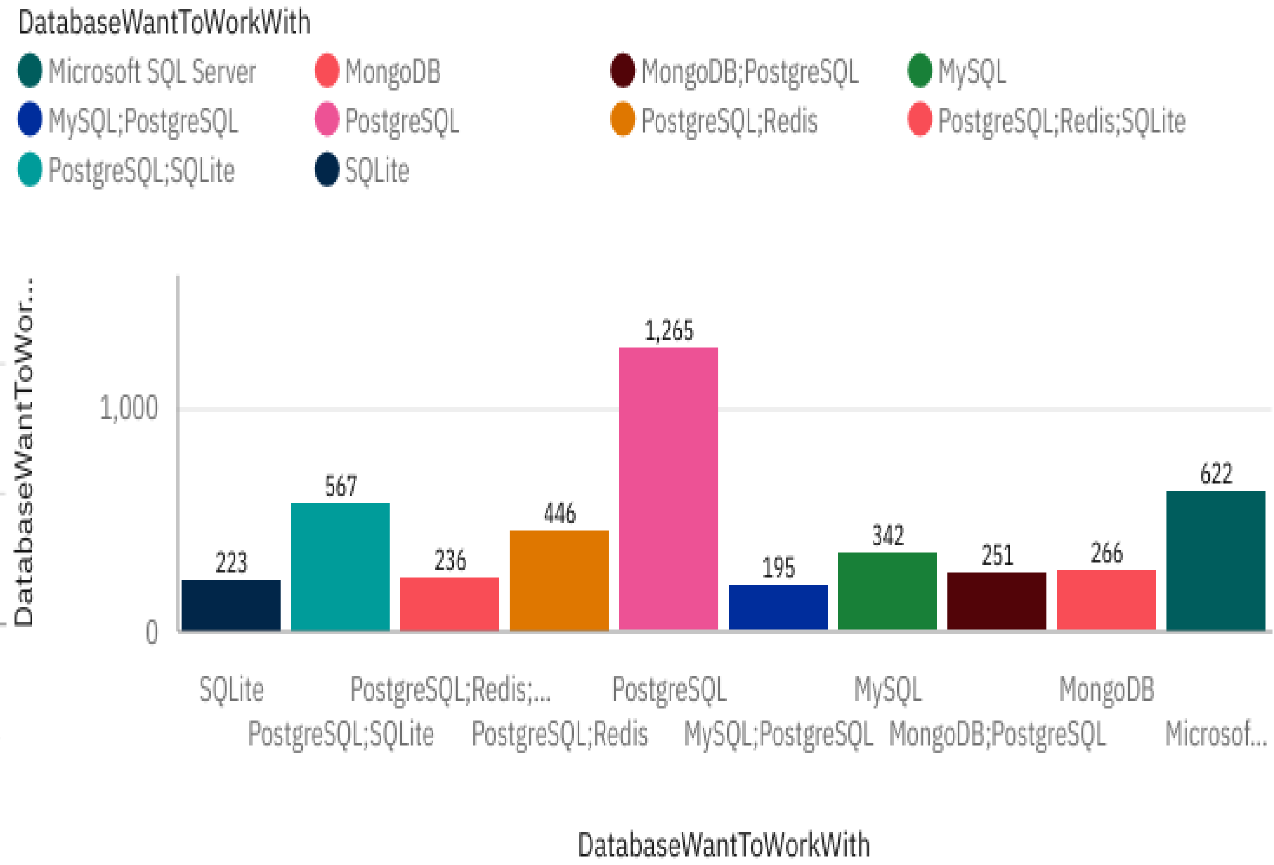
Current Year

Top 10 Databases Currently Used



Next Year

Top 10 Databases Respondents Want to Work With



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- A small number of widely used relational databases currently make up the majority of database usage, highlighting continued reliance on developed and tested data management systems.
- Due to a variety of data storage demands, a blend of relational and non-relational (NoSQL) databases is used both now and in the future.
- Compared to current usage, future demand indicates a greater interest in cloud-friendly and scalable databases.

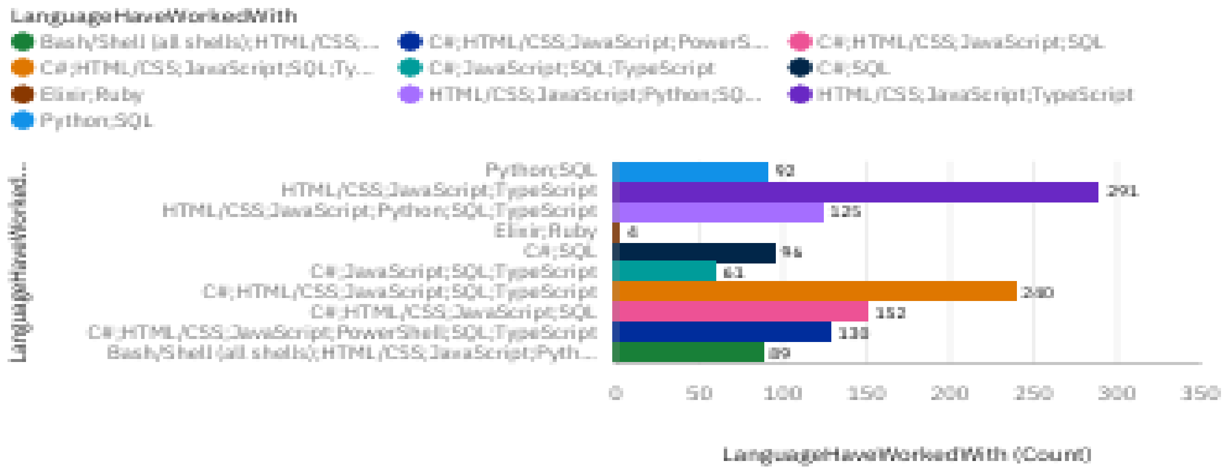
Implications

- Data professionals must maintain their knowledge with widely used relational databases while learning more about NoSQL and cloud-based alternatives.
- In order to handle both structured and unstructured data requirements, organizations may need to implement flexible data strategies.
- To prepare for changing data management trends, training and education programs must incorporate current database technologies.

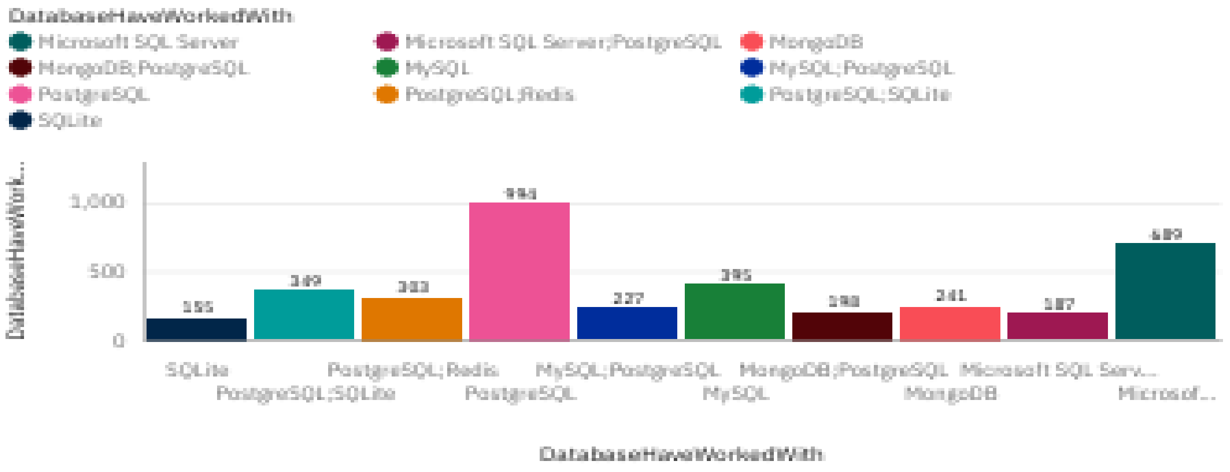
DASHBOARD TAB 1

Current Technology Usage

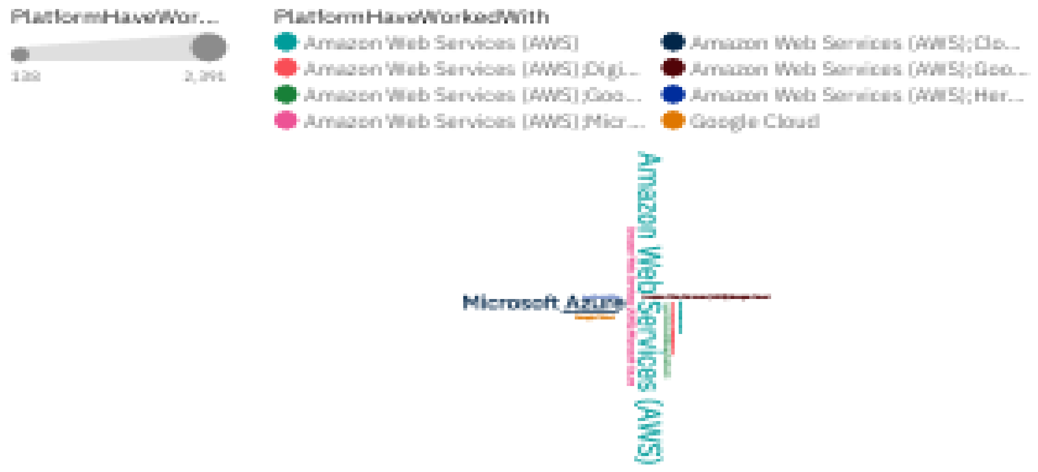
Top 10 Programming Languages Currently Used



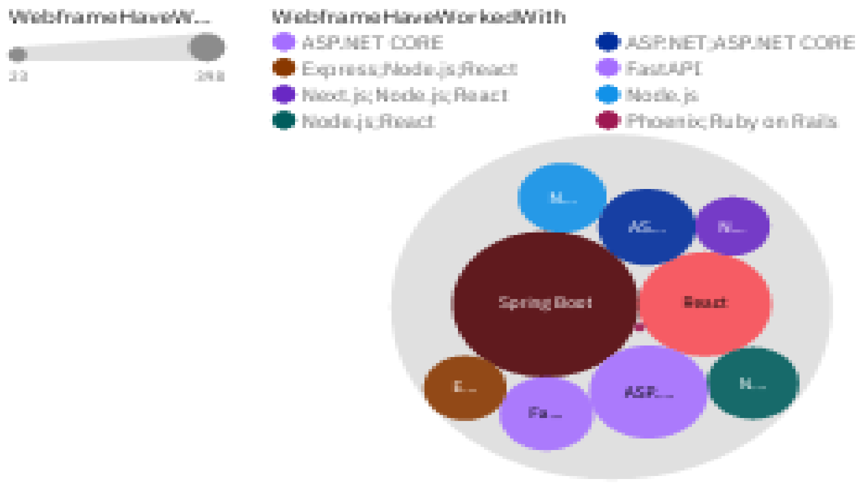
Top 10 Databases Currently Used



Top 10 Platforms Currently Used



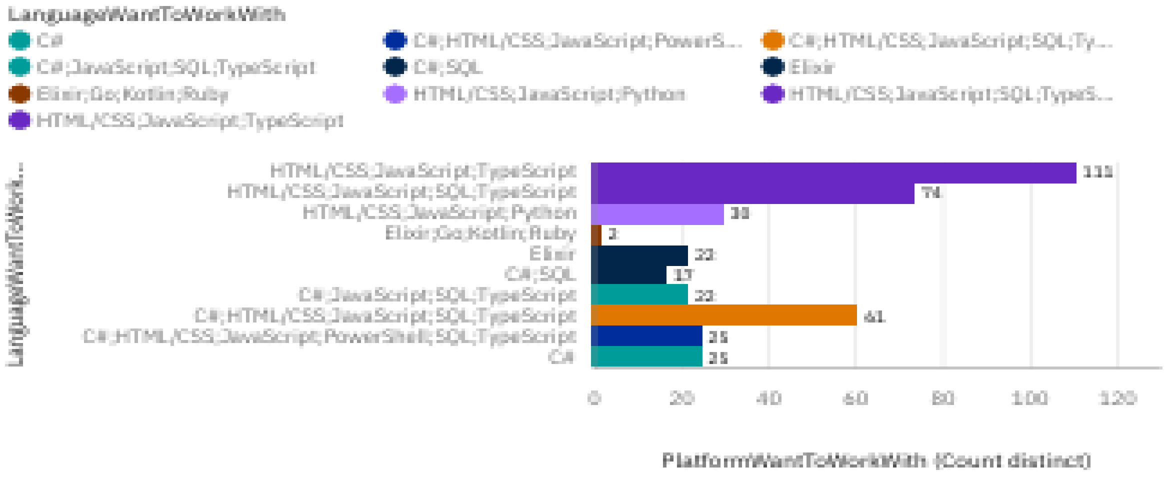
Top 10 Web Frameworks Currently Used



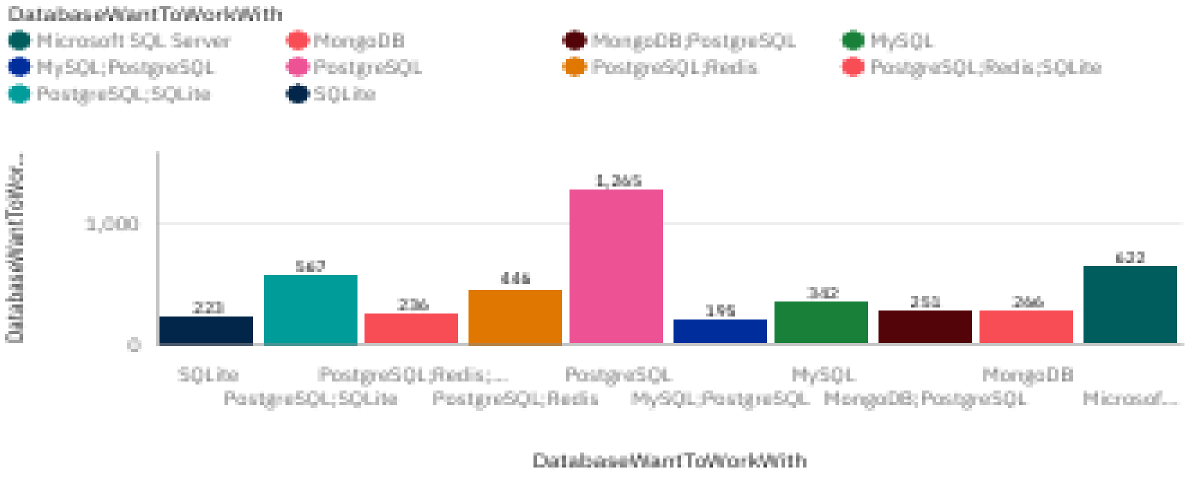
DASHBOARD TAB 2

Future Technology Trend

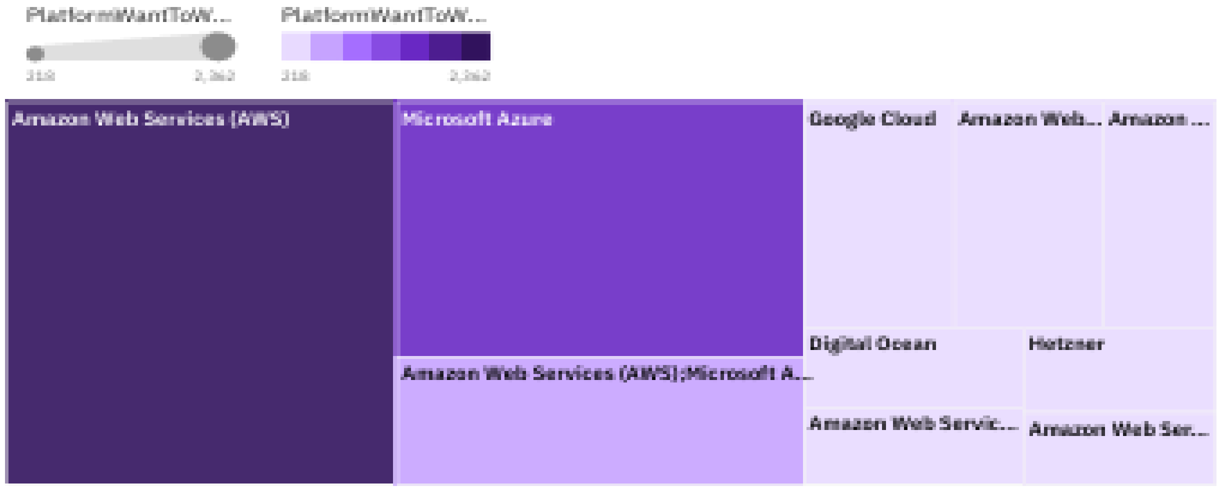
Top 10 Programming Languages Respondents Want to Work With



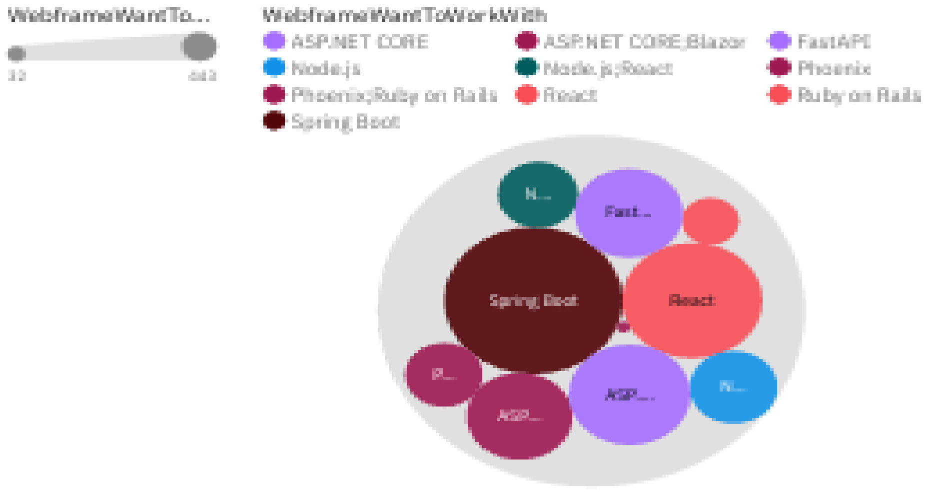
Top 10 Databases Respondents Want to Work With



Top 10 Platforms Respondents Want to Work With



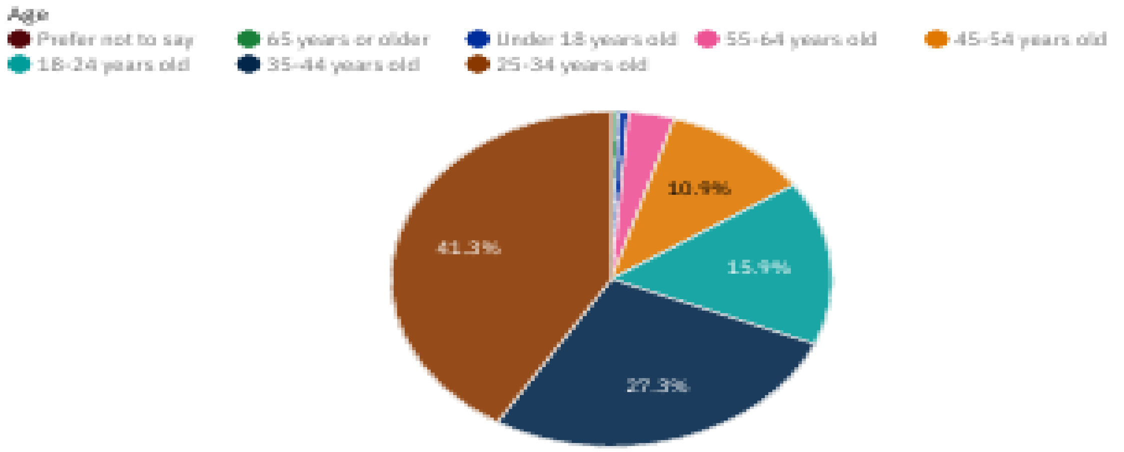
Top 10 Web Frameworks Respondents Want to Work With



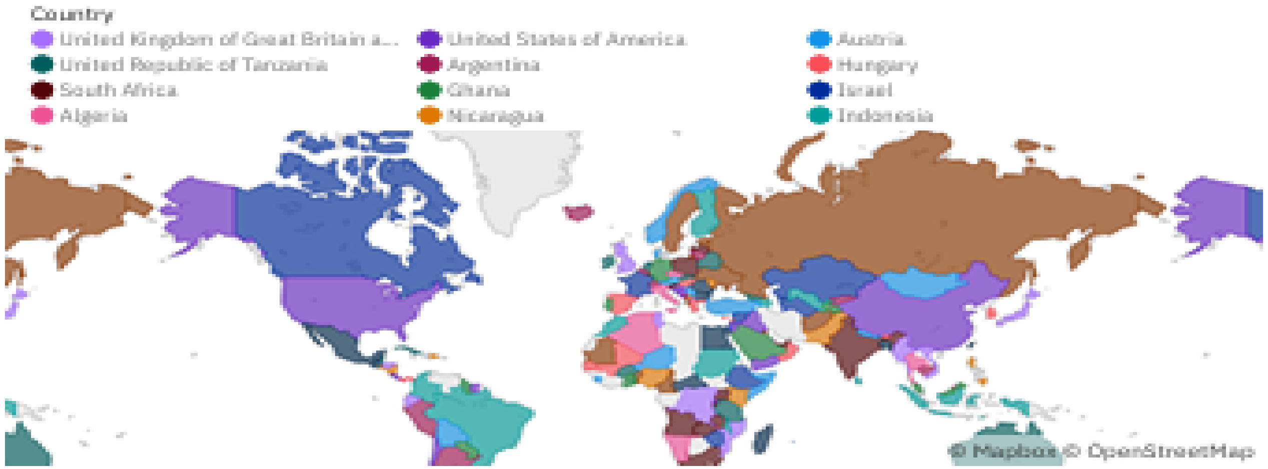
DASHBOARD TAB 3

Demographics

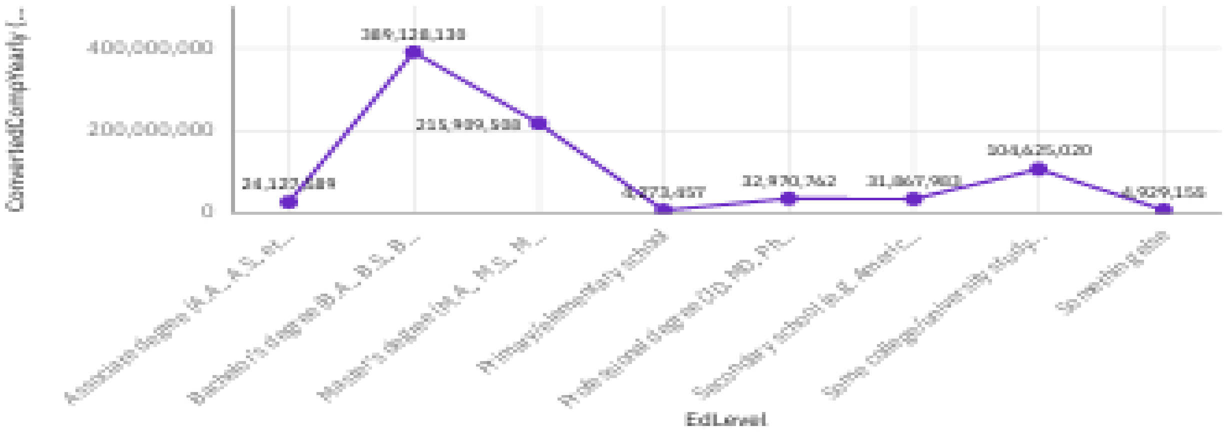
Respondent Distribution by Age



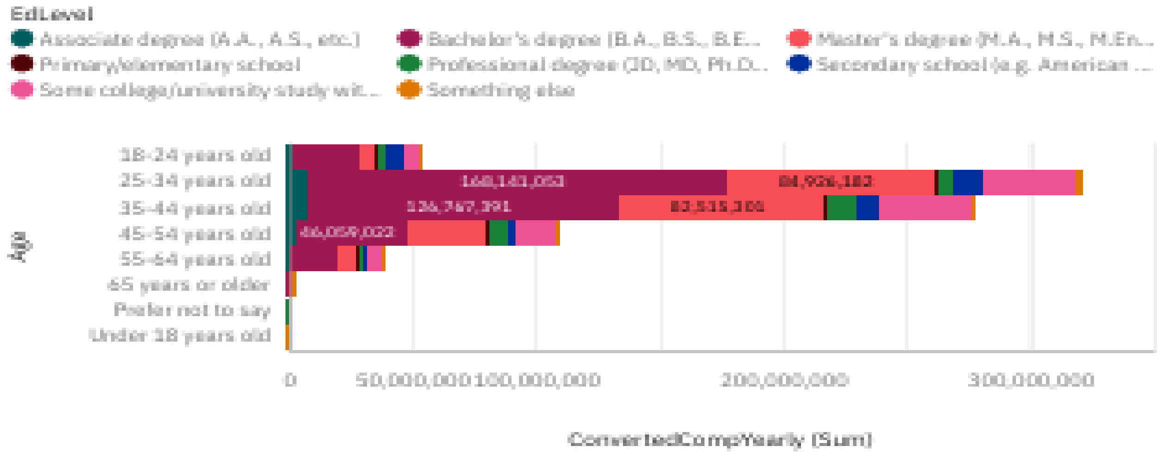
Respondent Count by Country



Respondent Distribution by Formal Education Level



Respondent Count by Age and Education Level



DISCUSSION

- Current usage is concentrated around a small number of widely adopted programming languages, databases, platforms, and web frameworks.
- Future trends indicate growing interest in emerging and modern technologies across programming languages, databases, platforms, and web frameworks.
- Most respondents fall within the early to mid-career age range, suggesting a workforce actively developing and expanding technical skills.

OVERALL FINDINGS & IMPLICATIONS

Findings

- A small number of reputable and widely used technologies from programming languages, databases, platforms, and web frameworks dominate the adoption of technology today.
- Future technological trends show a shift in the demand for skills as people become more interested in current, scalable, cloud-based solutions.
- A globally dispersed and rapidly developing technological workforce is suggested by the diverse mix of demographics.

Implications

- To stay relevant in a constantly changing technological context, professionals must pursue ongoing education and upskilling.
- To stay competitive, companies should match their technology, hiring, and training strategies with new trends.
- These insights can be used by educational institutions to create courses which find a balance between future technical demands and industrial standards.

CONCLUSION

- The project successfully used IBM Cognos Analytics to visualize and analyze technology survey data through interactive dashboards.
- An analysis of current and future technological trends showed that there is a profound reliance on popular tools in addition to an increasing interest in new technologies.
- A diverse respondent base with a range of age groups, levels of education, and regional representation was made clear by demographic insights.
- Overall, the results show that current industry practices and projected future technological demands are in line.